UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Triangular Arrays and Lindeberg's CLT

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1 Recap of STAT 553 & 575

Definition 1 (Convergence in distribution)**.** $X_n \xrightarrow{\text{d}} X$ *if* $F_{X_n}(t) \to F_X(t)$ *as* $n \to \infty$ *for every* $t \in \mathbb{R}$ *at which* F_X *is continuous.*

Remark. $\{(-\infty, t] : t \in \mathbb{R}\}$ *is generator for Borel* σ*-algebra, so CDF uniquely characterizes distribution.*

Definition 2 (Characteristic function (CF))**.**

$$\varphi_{X_n}(t) = \mathbb{E}[e^{itX_n}]$$

It is the Fourier transform of the law of X_n

Theorem 1. $[X_n \overset{d}{\Rightarrow} X] \iff [\varphi_{X_n}(t) \to \varphi_X(t),\ \forall t]$

Exercise 1. Show the $(\Rightarrow)$ direction in the previous Theorem.
Hint: Skorokhod representation + DCT.

Theorem 2 (Levy's)**.** Assume $\lim_{n\to\infty} \varphi_{X_n}(t)$ exists and equals $\varphi(t)$, then

(a) If $\varphi(t)$ is discontinuous at 0, then X_n does not converge in dist.

(b) If $\varphi(t)$ is continuous at 0, then $\varphi(t)$ is the CF of a r.v. X and $X_n \overset{d}{\Rightarrow} X$.

1.1 Convergence Determining Classes

Definition 3. A class of functions $\mathcal{C}$ is called *convergence determining* if the following holds:

$$X_n \xrightarrow{d} X \iff \mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)], \quad \forall f \in \mathcal{F}.$$

Examples:

1. $\mathcal{C}_b^k$: all $f : \mathbb{R} \to \mathbb{R}$, bounded with bounded $1^{st}, 2^{nd}, \ldots, k^{th}$ derivative.
2. $\mathcal{C}_b$: the set of all $\mathbb{R} \to \mathbb{R}$ continuous & bounded functions.
3. $\mathcal{C}_b^{Lip}$ the set of all $\mathbb{R} \to \mathbb{R}$ Lipschitz & bounded functions.
   Recall: $f : \mathbb{R} \to \mathbb{R}$ is Lipschitz if $\exists L > 0$ s.t.
   $$|f(x) - f(y)| \leq L|x - y| \quad \forall x, y \in \mathbb{R}.$$
4. $\mathcal{C} = \{\cos t, \sin t\}$.

Exercise 2. Show that the class of indicator functions, $1_{(-\infty,t]}$ can be approximated with continuous functions.

2 Central Limit Theorems

Recall the classical CLT:

Theorem 3 (Classical CLT)**.** *If $X_1, \ldots, X_n$ is an iid sequence with $\mathbb{E}[X_i] = \mu$, $\mathrm{Var}(X_i) = \sigma^2 < \infty$*
Then,
$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \xRightarrow{d} N(0, 1)$$

Exercise 3. *Remind yourself proof of Classical CLT.*

2.1 Lindeberg's CLT

The classical CLT applies to iid random variables. Going beyond, one main goal of the lecture is to prove Lindeberg's CLT.

Theorem 4. ***Lindeberg's CLT:*** Let $X_1, \ldots, X_n$ be independent with $\mathbb{E}[X_i] = \mu_i$, $\mathrm{Var}(X_i) = \sigma_i^2 < \infty$, and set

$$S_n^2 = \sum_{i=1}^{n} \sigma_i^2.$$

Then,

$$\frac{1}{S_n} \sum_{i=1}^{n} (X_i - \mu_i) \overset{d}{\Rightarrow} N(0,1),$$

if the Lindeberg condition holds:

$$\forall \varepsilon > 0, \ \sum_{i=1}^{n} \mathbb{E}\Big[|\frac{X_i - \mu_i}{S_n}|^2 \mathbb{1}\{|\frac{X_i - \mu_i}{S_n}| > \varepsilon\} \Big] \to 0,$$

as $n \to \infty$.

Triangular Array: Roughly speaking, sum of many small r.v.s converges to normal. Consider the array of numbers:

$$\begin{array}{llllll} X_{11} & \ldots & X_{1n_1} & & & \\ X_{21} & \ldots & \ldots & X_{2n_2} & & \\ \vdots & & & & & \\ X_{i1} & \ldots & \ldots & \ldots & X_{in_i} & \end{array} \qquad \downarrow \text{ more numbers}$$

(More numbers as $n \to \infty$)

Example: $\mathrm{Bin}\big(n, \frac{1}{2} + \frac{1}{n}\big) = X_1 + \cdots + X_n$, where $X_i \sim \mathrm{Bern}\big(\frac{1}{2} + \frac{1}{n}\big)$ iid.

Definition 4. *(Triangular Array Condition [TAC])*

1. *For each row i, $X_{i1}, \ldots, X_{in_i}$ are independent (not necessarily identically distributed). Different rows need not be independent.*
2. $\mathbb{E}(X_{ij}) = 0$ *for every $j \le n_i$ and row $i \ge 1$.*
3. $\sum_{j \le n_i} \mathbb{E}(X_{ij}^2) = 1$ *for each row i. (Row variance is* 1*.)*

Suppose $\{X_{ij}\}$ satisfy TAC & that Lindeberg condition holds:

$$\forall\, \varepsilon > 0: \quad \sum_{j=1}^{n_i} \mathbb{E}\Big[(X_{ij})^2\, 1_{\{|X_{ij}|>\varepsilon\}}\Big] \;\to\; 0 \quad \text{as } i \to \infty.$$

Then, Lindeberg's theorem states that $S_i \overset{d}{\Rightarrow} N(0,1)$ where $S_i = \sum_{j=1}^{n_i} X_{ij}$.

Observations:

$$\begin{aligned} X_{ij}^2 &\le \varepsilon^2 + X_{ij}^2 \mathbf{1}_{\{|X_{ij}|>\varepsilon\}} \\ \mathbb{E}[X_{ij}^2] &\le \varepsilon^2 + \mathbb{E}[X_{ij}^2 \mathbf{1}_{\{|X_{ij}|>\varepsilon\}}] \\ \mathbb{E}[X_{ij}^2] &\le \varepsilon^2 + \sum_{j=1}^{n_i} \mathbb{E}[X_{ij}^2 \mathbf{1}_{\{|X_{ij}|>\varepsilon\}}] \end{aligned}$$

Above is true for all j. Therefore,

$$\max_{j\le n_i} \mathbb{E}[X_{ij}^2] \le \varepsilon^2 + \sum_{j=1}^{n_i} \mathbb{E}[X_{ij}^2 \mathbf{1}_{\{|X_{ij}|>\varepsilon\}}],$$

and $\lim_{i\to\infty} \max_{j\le n_i} \mathbb{E}[X_{ij}^2] = 0$

Fact: For a triangular array, $n_i \to \infty$ must hold:

$$1 = \sum_{j=1}^{n_i} \mathbb{E}[(X_{ij})^2] \le n_i \cdot \underbrace{\max_{j\le n_i} \mathbb{E}[(X_{ij})^2]}_{\longrightarrow 0} \qquad \therefore n_i \to \infty.$$

Definition 5 (Uniformly Asymptotically Negligible (UAN))**.**

$$\lim_{i\to\infty} \max_{j\le n_i} \mathbb{P}[|X_{ij}| > \varepsilon] = 0.$$

Thus, along each row, the X_{ij} tends to 0 in probability uniformly.

Theorem 5 (Lindeberg⇒UAN)**.**

$$\mathbb{P}(|X_{ij}| > \varepsilon) \le \frac{1}{\varepsilon^2}\mathbb{E}[X_{ij}^2] \le \frac{1}{\varepsilon^2} \max_{j\le n_i} \mathbb{E}[X_{ij}^2]$$

$$\Rightarrow \max_{j\le n_i} \mathbb{P}(|X_{ij}| > \varepsilon) \le \frac{1}{\varepsilon^2} \max_{j\le n_i} \mathbb{E}[X_{ij}^2] \to 0$$

Theorem 6. *(Feller)* Suppose that TAC holds, and that the sequence is UAN. Then $S_i \xrightarrow{d} N(0,1) \iff$ it satisfies Lindeberg condition.

The proof can be found in *(Billingsley or Kallenberg).*

Assuming Lindeberg's CLT, we prove ordinary CLT.

Proof of Ordinary CLT:

$$\frac{1}{\sigma\sqrt{n}}\sum_{j=1}^{n}(X_j-\mu)=\sum_{j=1}^{n}\frac{X_j-\mu}{\sigma\sqrt{n}}.$$

$$\mathbb{E}\left[\left(\frac{X_j-\mu}{\sigma\sqrt{n}}\right)^2\right]=\frac{1}{n}\mathbb{E}\left[\left(\frac{X_j-\mu}{\sigma}\right)^2\right]=\frac{1}{n} \qquad \longleftarrow \text{TAC.}$$

Next, we verify the Lindeberg's condition:

$$\begin{aligned}&\sum_{j=1}^{n}\mathbb{E}\Big[(\frac{X_j-\mu}{\sigma\sqrt{n}})^2\mathbb{1}\{|\frac{X_j-\mu}{\sigma\sqrt{n}}|>\varepsilon\}\Big]\\ =&n\mathbb{E}\Big[(\frac{X_1-\mu}{\sigma\sqrt{n}})^2\mathbb{1}\{|\frac{X_1-\mu}{\sigma\sqrt{n}}|>\varepsilon\}\Big]\\ =&\mathbb{E}\Big[(\frac{X_1-\mu}{\sigma})^2\mathbb{1}\{|\frac{X_1-\mu}{\sigma\sqrt{n}}|>\varepsilon\}\Big]\end{aligned}$$

Note that

$$\left(\frac{X_j-\mu}{\sigma}\right)^2\mathbb{1}\left\{\left|\frac{X_j-\mu}{\sigma}\right|>\varepsilon\sqrt{n}\right\}\le\left(\frac{X_1-\mu}{\sigma}\right)^2.$$

Furthermore,

$$(\frac{X_1-\mu}{\sigma})^2\mathbb{1}\left\{|\frac{X_1-\mu}{\sigma}|>\varepsilon\sqrt{n}\right\}\overset{n\to\infty}{\to}0$$

pointwise. Therefore, by Dominated Convergence Theorem (DCT),

$$\frac{1}{\sigma^2}\mathbb{E}\left[(X_j-\mu)^2 1_{\{|X_j-\mu|>\varepsilon\sigma\sqrt{n}\}}\right]\longrightarrow 0,$$

verifying Lindeberg's condition.

3 Proof of Lindeberg's CLT

Let $X_j := X_{ij}, \quad S = \sum_j X_{ij} \, (= S_i)$.

Let $Z_i, \ldots, Z_n$, be independent standard normal with $\mathbb{E}(Z_i) = 0$, $\text{Var}(Z_i) = \sigma_i^2$. We are going to 'replace' X_i with Z_i which match in the first two moments.

$$
\begin{aligned}
W_0 &= X_1 + \ldots + X_n \\
W_1 &= Z_1 + X_2 + \ldots + X_n \\
W_i &= Z_1 + Z_2 + \cdots + Z_i + X_{i+1} + \ldots + X_n \\
W_n &= Z_1 + \cdots + Z_n \sim N(0,1), \quad \sum \sigma_i^2 = 1 \quad \text{(due to TAC)}
\end{aligned}
$$

Want $W_n \overset{d}{\Rightarrow} W$, it suffices to show

$$\mathbb{E}[f(W_n)] \to \mathbb{E}[f(W)] \quad \text{for every } f \in \mathcal{C}_b^{(3)}.$$

$$|\mathbb{E}f(W_0) - \mathbb{E}f(W_n)| = \left| \sum_{i=0}^{n-1} \mathbb{E}f(W_i) - \mathbb{E}f(W_{i+1}) \right| \overset{\Delta\text{-inequality}}{\leq} \sum_{i=0}^{n-1} |\mathbb{E}f(W_i) - \mathbb{E}f(W_{i+1})|$$

Taylor: Assume $y < x$

$$f(x) = f(y) + f'(y)(x-y) + \frac{f''(y)}{2}(x-y)^2 + \cdots + \frac{f^{(n)}(\xi)}{n!}(x-y)^n \quad \text{for some } \xi \in (y, x)$$